

Plants vs Zombies AA- Markdown Planning Sheet

1. Decomposition

The tinier problems that we need to dissect to make the game work:

1. **Plant Class** and their *attributes* and *methods*:
[name, health, damage ; attack(), take_damage()]
2. **Zombie Class** and their *attributes* and *methods*:
[name,health,damge ; attack(), take_damage(),move()]
3. Create 2 Plant objects with different damage values and 1 Zombie object.
4. Turn loop that repeats until either side wins.
5. Plant's *attacking* phase. (*Each of the 2 living plants attack in order*)
6. Zombies's *attacking* phase. (*Close the distance by 1, or attack if distance at 0*)
7. Who will **Zombie** eat? (*First living plant*)
8. Check for the **winner**. (*Zombie health ≤ 0 , Both Plants health ≤ 0*)
9. Printing *actions*, *health values*, and *distance* of the **Zombie**.

2. Pattern Recognition

Name actions or checks that happen again each turn:

- Each living plant attacks the **Zombie** once
- Check if the **Zombies's** health is 0 or less
- Check if the **Zombies's** distance is greater than 0
- Check if both plants's health is 0 or less
- Both classes use the same kind of methods [*attack()*, and *take damage()*]

3. Abstraction

The essential information each character needs:

- **Plant**: name, health, damage (*Position and a Graphic is not needed.*)
- **Zombie**: name, health, damage, distance
- Aspects not needed: Graphics, Menus, Keyboard input

4. Algorithm Design

Game steps in order:

1. Create the two plants, **Potato** (Health 65, Damage 15), and **Tomato** (Health 70, Damage 10), and the zombie, **Kaiju**. (Health 250, Damage 30, Distance 6)
2. Start at **Turn 1**.
3. Print the current turn number.
4. For each plant, in order:
If the plant's health is > 0 , it attacks Kaiju and Kaiju takes damage.
If Kaiju's health is 0 or less, print "Plants win." and stop.
5. If Kaiju's distance is > 0 , it moves one step closer.
6. Otherwise, Kaiju attacks the first plant whose health is above 0.
7. If both plants's health is ≤ 0 , print "Kaiju wins" and stop.
8. Print the health of both plants and of Kaiju at the end of the turn.
9. Add 1 to the turn number and go back to step 3.

THE GAME STOPS WHEN PLANTS OR ZOMBIES WIN.
